



Scheme/ Answer Key for Valuation

Scheme of evaluation (marks in brackets) and answers of problems/key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER B. TECH DEGREE EXAMINATION, DECEMBER 2021

Course Code: PHT 100

Course Name: ENGINEERING PHYSICS A

PART A (MODULE I TO V)

Answer all questions, each carries 3 marks.

Marks

- | | | |
|----|--|-----|
| 1 | Definition --- (1mark)
Explanation ---- (1mark)
Expression for Q ----- (1mark) | (3) |
| 2 | $f = 1/2l\sqrt{T/m}$ ----- 1 mark
Substitution ----- 1mark
Answer – 700Hz ----- 1mark | (3) |
| 3 | Actual path difference -- $2\mu t \cos r \pm \lambda/2$ --- 1 mark
At the centre $t = 0$, path difference = $\lambda/2$ --- 1 mark
For $I_{\min}$, path difference = $(2n + 1) \lambda/2$ --- 1 mark | (3) |
| 4 | Definition ---- 2 marks
Expression with terms --- 1 mark | (3) |
| 5 | Uncertainty principle $\Delta E \Delta t \geq \hbar/2$ (1mark),
Substitution (1 mark),
Answer with unit $\Delta v \geq 7.98 \text{ MHz}$ (1 mark), | (3) |
| 6 | Larger surface to volume ratio ----- (1 mark),
Explanation (2 marks) | (3) |
| 7 | Statement: 2 marks, Mathematical expression: 1 mark | (3) |
| 8 | Statement --- 1 mark, Physical significance --- 2 marks | (3) |
| 9 | Definition of cooper pairs 1 mark
Role in superconductivity 2 marks | (3) |
| 10 | Definition of photo detector ---- 2 marks, Two examples --- 1mark | (3) |



PART B (MODULE I TO V)

Answer one full question from each module, each question carries 14 marks

- 11 (a) Mention the forces acting on the system and obtain differential equation ---- (10)
 -- 3 marks
 Assuming solution, $x = A \sin(pt - \Theta)$ ----- 1 mark
 Derivation to find amplitude A ----- 3 marks
 Finding phase Θ --- 1/2 mark
 Final solution ----- 1/2 mark
 Amplitude resonance ----- 2 marks
- (b) Formula--- $\omega_0 = \sqrt{C/m}$ (0.5 mark), substitution (1 mark) answer with unit ---
 -(0.5 mark); Angular frequency without damping=100 rad/s (4)
 Formula--- $\omega = \sqrt{\omega_0^2 - k^2}$ (0.5 mark), substitution (1 mark) and answer with
 unit---(0.5 mark); Angular frequency with damping=99.99 rad/s
- 12 (a) Figure.....1mark; Explanation.....2marks; Equation of Resultant (10)
 force....1 mark; Derivation of Differential eqn....3 marks
 General wave equation.....1 mark; Prove that $v = \sqrt{T/m}$ ----1mark;
 Frequency $f = (1/2l) \sqrt{T/m}$ 1 mark
- (b) i. Amplitude = $3 \times 10^{-3} \text{m}$ -----1mark; ii. Frequency = $1.34 \times 10^{13} \text{Hz}$ --formula
 with answer.... 1mark; iii. wavelength = $22.4 \times 10^{-6} \text{m}$ formula with (4)
 answer ...1mark; iv. wave velocity = $3 \times 10^8 \text{ m/s}$ --- formula with answer
1mark
- 13 (a) Figure-1 mark, (10)
 Explanation-2 marks.
 Derivation to find β -- 3 marks
 Explanation of expt. to find d --- 2 marks
 Derivation to find d --- 2 marks
- (b) Radius of curvature, $R = 0.879 \text{m}$
 Formula $R = D_{20}^2 - D_{10}^2 / 4n\lambda$ (1mark); Substitution and calculation (2marks); (4)
 answer with unit (1 mark)



- 14 (a) Fig---1 mark, grating and grating element---2 marks, derivation---3 marks, (10)
Rayleigh's criterion- statement and explanation with fig---4 marks
(b) $\sin \theta = N \lambda$ --- 1 mark
Substitution & finding the value of N --- 2 marks
To find $N_0 = N \times \text{Width} = 9420 \text{ lines}$ --- 1 mark (4)
- 15 (a) Time dependent equation (10)
Equation for ψ --- 1 mark
Equation for $E\psi$ --- 1 mark
Equation for $P^2\psi$ --- 1 mark
Expression for energy ---- $\frac{1}{2}$ mark
Obtaining the Schrodinger equation --- 2 marks
Time independent equation
Express ψ as product of two functions --- 1 mark
To find $\frac{\partial \psi}{\partial t}$ --- 1 mark
To find $\frac{\partial^2 \psi}{\partial x^2}$ --- 1 mark
Obtaining the final Schrodinger equation --- 1.5 marks
(For writing Schrodinger equation without derivation ---- 1 mark each)
- (b) Uncertainty principle equation - $\Delta x \Delta p \geq \hbar/2$ (1mark) (4)
substitution and calculation (2 marks), Result with unit (1mark)
 $\Delta v = 2.898 \text{ km/s}$
- 16 (a) Properties---2 marks each, 4 applications---4 marks (10)
(b) Quantum Confinement----4 marks (4)



17

- (a) Defining the terms magnetisation, magnetic flux density, magnetic permeability, relative permeability and susceptibility. (1.5 marks each). (10)

Obtaining the relation between relative permeability and susceptibility- 2.5 marks.

- (b) $M = \chi H$ Formula ...1 mark ; $M = -4.92 \text{ A/m}$ substitution and answer.... 1 mark, $B = \mu_0 (H + M)$ Formula1 mark, $B = 0.75 \text{ Wb/m}^2$ substitution and answer1 mark (4)

18

- (a) Derivation of the four equations with complete steps starting from basic laws---- 3+3+3+4 Marks { 3 marks of 18 (b) added to 18(a) } (13)

(b) $E/H = \sqrt{\frac{\mu_0}{\epsilon_0}}$ --- 1 mark OR

$E/H = 377 \text{ Ohms}$ --- 1 mark

(1)

[As plane wave solution is not specifically mentioned in the syllabus, 3 marks from 18 (b) may be added to 18 (a)]

19

- (a) Describing the phenomenon of superconductivity----- 2 marks. (8)

Defining critical temperature and critical magnetic field---2 marks each.

Any 4 applications of superconductors ----- 0.5 marks each

- (b) 2 graphs ----- 2 Marks

Any 4 differences ----- 1 Mark each

(6)

20

- (a) Definition of acceptance angle and numerical aperture – 2 marks each (10)

Labelled diagram ...1 mark; Explanation1mark; Derivation of the relation of NA in terms of refractive indices....4 marks

- (b) $NA = \sqrt{n_1^2 - n_2^2}$ Formula.... 1 mark

$n_1 = 1.5599$ substitution and answer 1 mark

Acceptance angle $= \sin^{-1} NA = 30.497^\circ / 30^\circ 29'$; (Formula.... 0.5 mark, substitution and answer with unit0.5 mark)

$$\sin \theta_c = \frac{n_2}{n_1} \quad \text{Critical angle } \theta_c = \sin^{-1} \frac{1.475}{1.5599} = \sin^{-1} 0.9456 = 71^\circ \quad (4)$$

(Formula.... 0.5 mark, substitution and answer with unit0.5 mark)
